Gridiron Glove

EXECUTIVE SUMMARY: GRIDIRON GLOVE

Problem

Youth football is a very large activity in America with millions of kids playing every fall, and wide receivers are one of the most technique dependent positions on the field. Catching the ball with the hands instead of the body, hand position, and timing of the finger close are all things that can be taught but only if somebody is watching and giving feedback. The problem is that most youth and middle school football programs are run by volunteer coaches who have a full roster of kids, limited practice time, and often no specific background in receiver play. High schools and colleges can use video analysis software and have assistant coaches for each position group, but youth programs mostly can not afford this. As a result bad habits like body catching and late hand closure are formed early and stick with players for years. Coaches we talked to (about 14 so far, mostly from rec leagues and two middle schools near our university) told us this is one of their biggest frustrations, and several said they would pay for a tool that tells them what to fix.

Solution

Gridiron Glove is a training glove with embeded sensors paired with a mobile application. The glove has pressure sensors in the palm and fingertips plus a small motion sensor on the back of the hand. When a player catches a ball the glove records the contact pattern and the timing and the app converts that into a catch technique score from 0 to 100. The score breaks down into sub scores such as hand placement, finger closure timing, and whether the ball was secured with hands only or trapped to the chest. After a practice session the app shows the coach a simple team dashboard, ranks players, and suggests specific drills from a library tied to each weakness. The coach does not need to review video or understand biomechanics. Our software is designed to be simple enough to use on a phone at the field while kids are doing reps.

Our prototype currently works with one glove size and a wired connection to a small battery pack and our scoring is based on rules we developed with advice from two coaches rather than a large data set. Wireless and multiple sizes are on the roadmap.

Market

According to widely cited industry figures there are several million youth tackle and flag football players in the United States and hundreds of thousands of youth teams. If even a small fraction of these teams bought a set of gloves, this would be a meaningful market. We estimate the total addressable market at over one billion dollars by multiplying the number of players by an assumed price, though we realize that not every player needs their own glove. Our more realistic serviceable market is youth leagues and middle school programs that already spend money on equipment, which we think is roughly 20 to 30 thousand programs. Our first target is the region around our university where we have relationships with a number of leagues, and then expanding state by state.

Business Model

We plan to sell a team kit of 6 gloves with the software included for the first year, priced around 450 dollars, and then charge an annual software subscription of about 120 dollars per team after that for the dashboard, drill library and updates. We also plan to sell single gloves to parents and individual players at roughly 99 dollars, and to sell to private receiver training camps. Our cost to build each glove at prototype stage is around 38 dollars in parts, and we believe this could fall at volume but we have not gotten manufacturing quotes yet. Longer term we see revenue from league wide licensing deals and from data insights, though the data side is early thinking.

Competition

There are video analysis platforms used by high school and college programs, but these are expensive and need cameras and staff time. There are also smart ball products and passing machines that help with throwing and catching drills but they do not measure what the hands are doing at the moment of the catch. Some general wearable sensor companies serve pro and college athletes at high price points. We are not aware of a product that targets youth coaches specifically with a low cost glove and drill suggestions. Our advantage is price, simplicity, and focus on a single position skill. Of course a bigger company could enter this space, and we have not filed any patents yet, which is a gap we know about.

Team

Marcus Delaney Otieno is our CEO and leads partnerships, he played receiver in high school and has coached a youth flag league for two seasons. Priya Raghunathan-Bell is CTO and leads hardware, she is a senior in electrical engineering with experience from a sensor research lab. Tobias Wendelstrom leads software and the mobile app and is studying computer science. Dani Kowalczyk-Reyes handles marketing and finance and is a business major who built our website. We do not yet have a team member with manufacturing experience or a formal advisor from the sports equipment industry, and we are looking to add one.

Milestones and Financials

So far we have defined the core concept, built a working prototype glove and a basic version of the app, registered the company, and launched a website. We have tested the prototype with a small number of players at two practices and collected early feedback which was positive though the sample is small and informal. We have spent roughly 2,300 dollars to date, funded by team members and a small university grant. Our next 12 months plan is to build 20 improved gloves, run a pilot with 4 to 5 youth teams, get manufacturing quotes, and file a provisional patent. We project revenue of about 18 thousand dollars in year one from pilot sales, growing to over 300 thousand dollars by year three, though these projections are based on assumptions about adoption that we still need to validate. We expect to be near break even at the end of year two if we keep the team lean.

What We Are Asking For

We are applying to the competition for funding, mentorship and visibility. We are requesting 25,000 dollars in prize or seed funding which we would use for improved prototype production, pilot costs, legal fees for the patent and for forming our legal entity, and basic testing. Just as important, we would like to be connected to mentors with experience in consumer hardware manufacturing, youth sports organizations, and early stage fundraising. We believe Gridiron Glove can make quality receiver coaching available to every kid regardless of their team's budget and we would be grateful for the chance to compete.